Question 1 from Quiz 4 asks us to find the first derivative of the distribution :

$$f(t) = \sin(t) \cdot u(\pi/2 - t).$$

Many students used the following approach :

$$f(t) = T_1(t) \cdot T_2(t),$$

$$T_1 = \sin(t) \quad \text{and} \quad T_2 = u(\pi/2 - t).$$

Apply product rule to distributions T_1, T_2 :

$$f(t) = T_1' \cdot T_2 + T_1 \cdot T_2' \quad \text{—} \; (1)$$

$$= \cos t \cdot u\left(\frac{\pi}{2} - t\right) + \sin(t) \cdot \delta(\pi/2 - t) \times (-1)$$

Note :

1. We did NOT prove the product rule in the course. In case you are using it, you must first prove it.

2. In fact, product rule (equation (1)) is NOT true in general, see below for a counter-example. Another counter-example at the very end.

3. For this particular question, differentiation by parts happens to work, and the final answer is correct. Why do we get the correct answer? See below for the special situation where product rule can be used.

I. Counter-example to Product Rule for Distributions.

Consider distributions:

$$f(t) = \begin{cases} t, & 0 \le t \le 1 \\ 0, & \text{else} \end{cases} = \mathbb{1}(0 \le t \le 1)\cdot t$$

$$g(t) = \begin{cases} t^2, & 0 \le t \le 1 \\ 0, & \text{else} \end{cases}$$

Clearly $g = f^2$. Let's check if the product rule works.

We need to compare:

$$g' \quad \text{and} \quad f\cdot f' + f'\cdot f = 2\cdot f'\cdot f$$

as distributions.

1. g':

$$\langle g', x\rangle = -\langle g, x'\rangle$$

$$= -\int_0^1 t^2 \cdot x'(t)\,dt$$

$$= -\int_0^1 \underbrace{t^2}_{u}\cdot d\underbrace{(x(t))}_{v}$$

$$= -\left\{ \left[t^2 x(t)\right]_0^1 - \int_0^1 x(t)\cdot 2t\,dt \right\}.$$

$$= -\left\{ x(1) - \int_{-\infty}^{+\infty} 2t\cdot\mathbb{1}(0\le t\le 1)\cdot x(t)\,dt \right\}.$$

$$\therefore\ g' = -\delta(t-1) + 2t\cdot\mathbb{1}(0\le t\le 1). \quad \text{—(2)}$$

2. Similarly, we can show:

$$f' = -\delta(t-1) + \mathbb{1}(0 \le t \le 1).$$

$$2ff' = 2\left[t \cdot \mathbb{1}(0 \le t \le 1)\right] \cdot \left[-\delta(t-1) + \mathbb{1}(0 \le t \le 1)\right]$$

$$= -2t\delta(t-1) + 2t \cdot \mathbb{1}(0 \le t \le 1)$$

$$= -2\delta(t-1) + 2t \cdot \mathbb{1}(0 \le t \le 1). \longrightarrow (3)$$

Clearly (2) & (3) are not the same.

Another technical hiccup is that $\delta(t-1) \cdot \mathbb{1}(0 \le t \le 1)$ is tricky to navigate because of the discontinuity at $t=1$. If the value of $\mathbb{1}(0 \le t \le 1)$ at $t=1$ is assumed to be 0.5 then $g' = 2ff'$. However, to be mathematically precise we are allowed to multiply $\delta(t-1)$ with only functions that behave well, such as Schwartz or at least continuous at $t=1$; $\mathbb{1}(0 \le t \le 1)$ is not such a well-behaved function.

My recommendation is to use $\langle T', x \rangle = -\langle T, x' \rangle$ in this course. This is fool-proof.

I have deducted 1 mark for using product rule without a proof.

II. When can I use the product rule?

<u>Theorem:</u> If $T(t) = T_1(t) \cdot \phi(t)$ where $\phi(t)$ is such that

$\phi(t) \times$ any Schwartz signal $=$ Schwartz signal, then product rule can be used:

$$T'(t) = T_1'(t)\cdot\phi(t) + T_1(t)\cdot\phi'(t).$$

Examples of such signals include

$\phi(t) =$ any polynomial in t, $\sin(\omega t)$, $\cos(\omega t)$, $e^{j\omega t}$.

In Quiz 4, Q.1 we have:

$$T(t) = u(\pi/2 - t)\cdot\sin(t).$$

and $u'(\pi/2 - t) = -\delta(t - \pi/2)$.

$\therefore$ From product rule:

$$T' = -\delta(t - \pi/2)\sin t + u\left(\tfrac{\pi}{2} - t\right)\cos t$$

$$= -\delta(t - \pi/2) + u\left(\tfrac{\pi}{2} - t\right)\cos t$$

since $\sin\frac{\pi}{2} = 1$

Proof of Theorem:

Suppose T_1 corres. to sequence of functions $f_1, f_2, \ldots$

Then $T_1(t)\cdot\phi(t)$ corrs. to:

$$f_1(t)\cdot\phi(t),\ f_2(t)\phi(t), \ldots$$

Note that $f_1\cdot\phi,\ f_2\cdot\phi, \ldots$ are all Schwartz.

Then, T' is associated with

$$(f_1\cdot\phi)',\ (f_2\cdot\phi)', \ldots$$

$= f_1\phi' + f_1'\phi,\ f_2\phi' + f_2'\phi, \ldots$

$= (f_1\phi',\ f_2\phi', \ldots)$

$+ (f_1'\phi,\ f_2'\phi, \ldots)$

All the functions in both sequences are Schwartz.

Let T_a corrs. to $f_1\phi', f_2\phi', \ldots$

Let T_b corrs. to $f_1'\phi, f_2'\phi, \ldots$

$$\langle T_a, x\rangle = \lim_{n\to\infty} \int_{-\infty}^{+\infty} f_n(t).\phi'(t).x(t)\,dt$$

$$= \langle T_1, \phi'.x\rangle$$

$$= \langle T_1\cdot\phi', x\rangle$$

$$\langle T_b, x\rangle = \lim_{n\to\infty} \int_{-\infty}^{+\infty} f_n'(t).\phi(t).x(t)\,dt$$

$$= \langle T_1', \phi.x\rangle$$

$$= \langle T_1'.\phi, x\rangle$$

$$\therefore\ T' = T_a + T_b$$

$$= T_1\cdot\phi' + T_1'.\phi.$$

III. Another Counter-Example

Consider $u(t)$ and $u^3(t) = u(t)\times u(t)\times u(t)$.

Irrespective of the value we assign to $u(0.5)$,

we see that

$$u(t) = u^3(t) \quad \forall\ t \neq 0.5$$

$\Rightarrow \langle u, x \rangle = \langle u^3, x \rangle$ for all Schwartz x.

$\Rightarrow$ u and u^3 are same (as distributions)

$\Rightarrow$ u and u^3 must have the same derivative

If the product rule holds here, then:

$$u' = 3 \cdot u' \cdot u^2 \quad \text{(as distributions)}$$

$$\delta(t) = 3 \cdot \delta(t) \cdot u^2(t)$$

$$\Rightarrow \langle \delta, x \rangle = \langle 3 \cdot \delta(t) \cdot u^2(t), x \rangle$$

$$x(0) = 3 \cdot u^2(0) \cdot x(0).$$

Even if we pick $u(0) = 0$ or 1 or 0.5 we see that $x(0) \neq 3 \cdot u^2(0) \cdot x(0)$.

$\Rightarrow$ This is a contradiction.